Logistics Performance Analysis

Identifying the Key Drivers of Late Deliveries in an E-commerce Network

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Business Problem

Context

E-commerce businesses rely heavily on fast and reliable delivery to maintain customer satisfaction and retention.

Problem

Recent delivery performance shows inconsistencies, with periods of increased late deliveries that may negatively impact customer experience and operational efficiency.

Objective

Identify the key drivers of late deliveries and determine whether delays are caused by:

- System-wide inefficiencies
- Regional challenges
- Specific underperforming sellers

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Dataset Overview

Data Source

- Brazilian E-commerce (Olist) dataset
- Public dataset representing real-world online retail operations

Key Tables Used

- Orders — purchase, delivery, and estimated dates
- Customers — location data (region/state)

Time Period

- Orders from **2016 to 2018**

Key Fields

- Order purchase timestamp
- Estimated delivery date
- Actual delivery date
- Seller ID
- Customer location (state/region)

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Key Business Questions

- How has **on-time delivery** performance changed over time?
- Are certain **regions** underperforming compared to others?
- Are specific **sellers** driving delivery delays?
- Is the issue **widespread** or concentrated among a **small group of sellers?**

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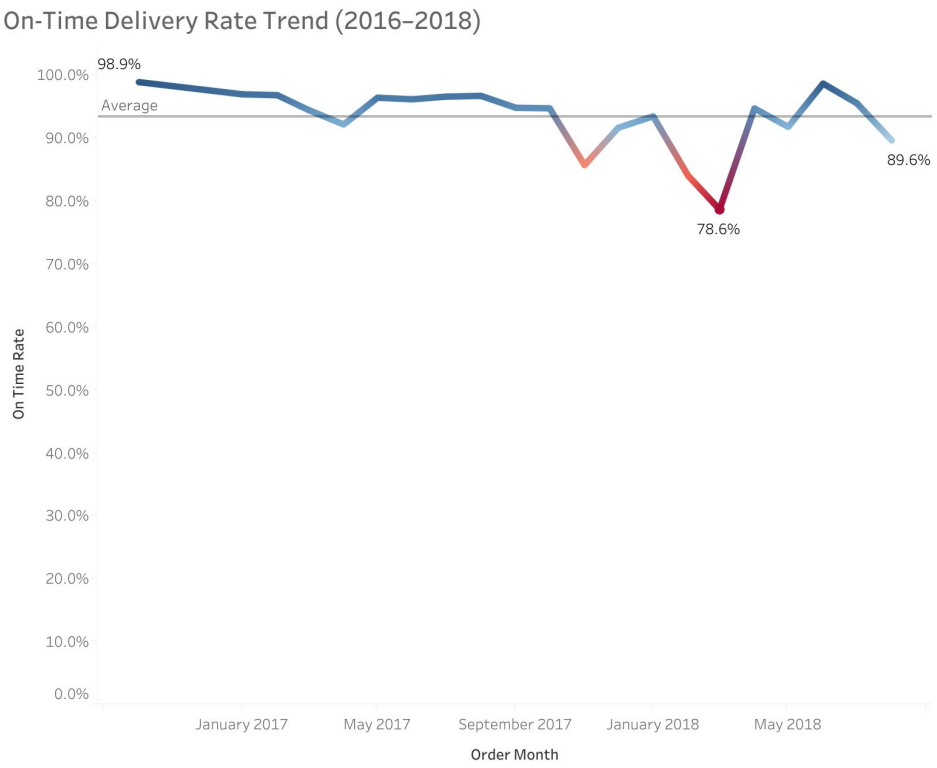
Delivery Performance is Strong but Inconsistent Over Time



- On-time delivery averages ~92%, indicating strong overall performance
- Performance shows **significant volatility**, with a low of ~78.6%
- Delivery reliability is **inconsistent over time**
- Suggests periods of **operational strain or inefficiency**

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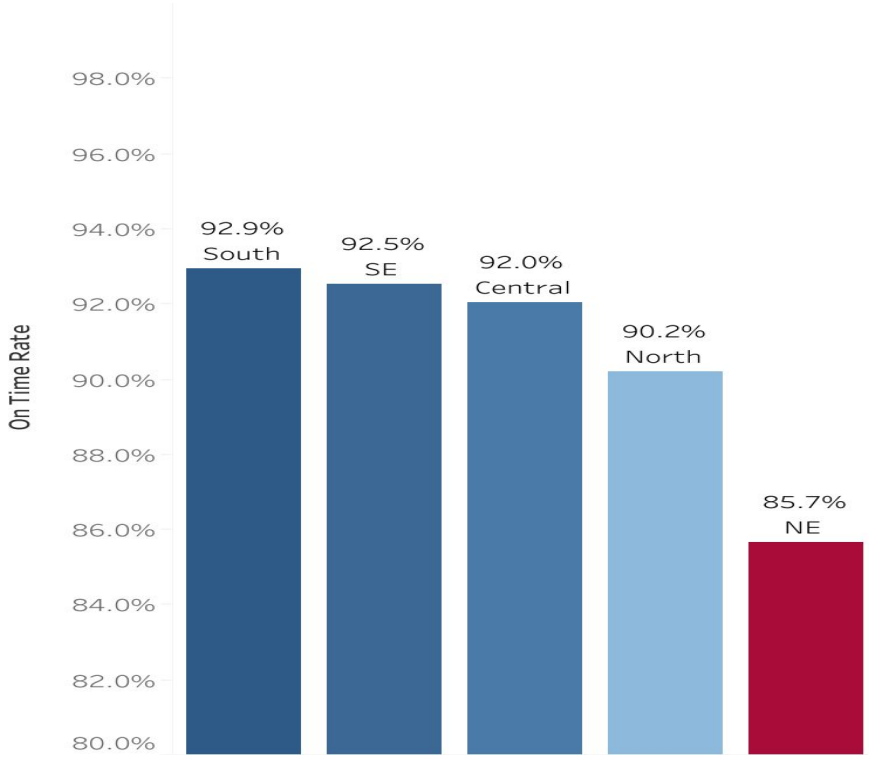
Delivery Performance Varies Significantly by Region



- Clear regional differences in delivery performance
- **Northeast is the lowest** at ~85.7% on-time
- Other regions consistently perform above 90%
- Indicates **geographic inefficiencies** in the delivery network

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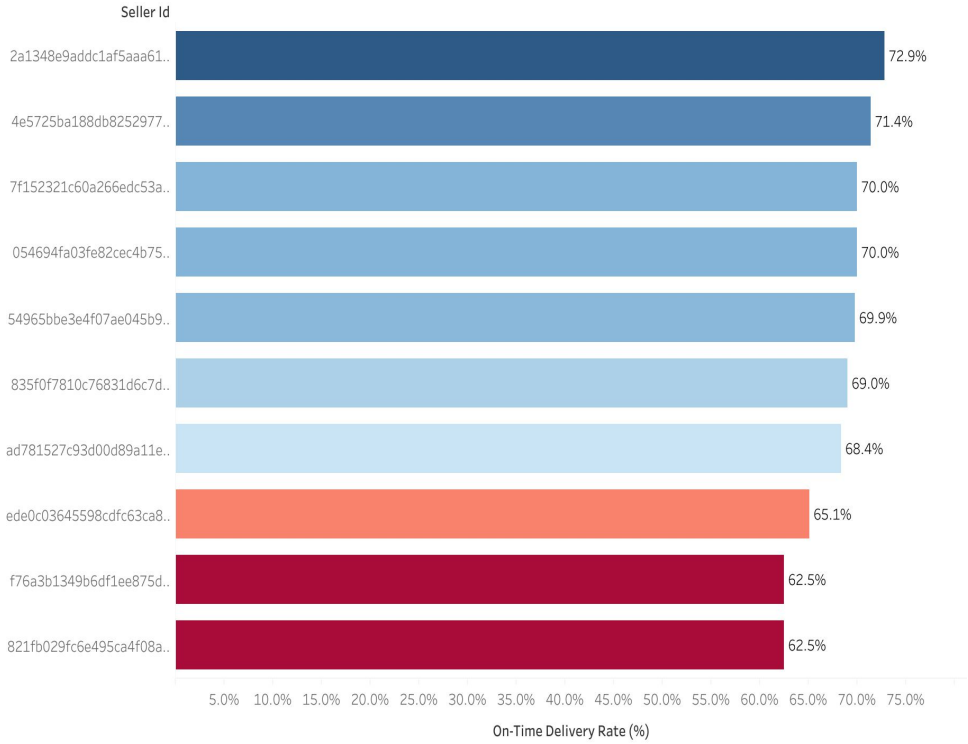
A Small Group of Sellers is Significantly Underperforming



- Bottom sellers operate at **~62%–73% on-time rate**
- Significantly below company average (~92%)
- Indicates **consistent underperformance**, not random delays
- **All worst performers are located in the Southeast**

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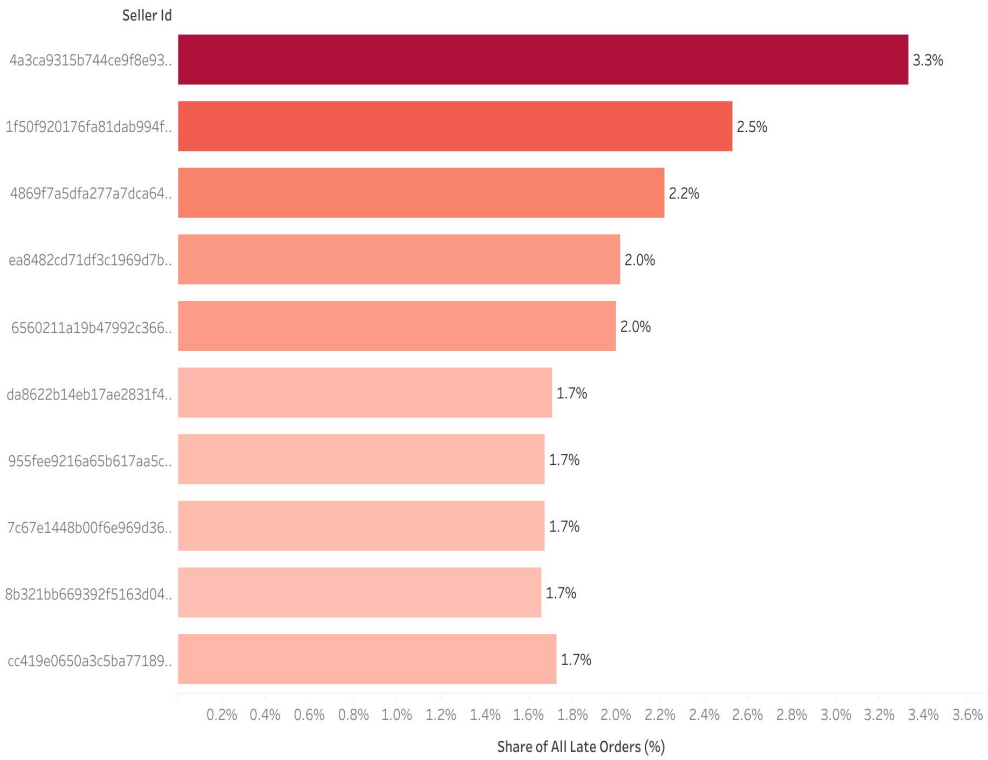
A Small Number of Sellers Drive a Disproportionate Share of Late Orders



- A small group of sellers contributes a **large share of late deliveries**
- Top sellers account for **~1.7%–3.3% each** of all late orders
- Delays are **highly concentrated**, not evenly distributed
- Indicates a **high-impact, targeted improvement opportunity**

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Key Insights

- Delivery performance is **strong overall (~92%) but inconsistent over time**
- Performance varies by region, with the **Northeast underperforming**
- A small group of sellers shows **significantly lower performance**
- These sellers are responsible for a **disproportionate share of late deliveries**

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Recommendations

1. Fix Underperforming Sellers

- Audit lowest-performing sellers and enforce minimum performance standards

2. Focus on High-Impact Sellers

- Target the small group of sellers driving the majority of delays

3. Strengthen Regional Logistics

- Address inefficiencies in the Northeast to improve delivery performance

4. Implement Performance Monitoring

- Track seller performance and flag underperformance early

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Appendix

Data Source

- Olist Brazilian E-commerce Dataset (2016–2018)

Key Notes

- On-Time Rate = % of orders delivered on or before estimated date
- Delay calculated as difference between actual and estimated delivery date
- Analysis performed at **order-level and seller-level aggregation**
- Only **delivered orders** included in analysis

Tools Used

- SQL (data extraction & transformation)
- Tableau Public (data visualisation & dashboarding)

Project Links

- Tableau Dashboard:

  https://public.tableau.com/app/profile/tony.deehan/viz/LogisticsAnalysis_17756624790930/Dashboard1

- Github Repository:

  https://github.com/tony-deehan/logistics-analysis

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